

Homework 4: Data Import and Enhanced Graphics

YOUR NAME HERE

2026-02-18

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# Load your libraries here
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Part I: Importing data

Go to Dr. D's Data page (<https://www.norcalbiostat.com/data/>) and download the following data sets.

- High School and Beyond
 - Crime Data
 - countyComplete
1. Read in the `High School and Beyond` data set. Use the R function `names` to get a list of variable names in this data set.
 2. Read in the `Crime Data` data set. Use R functions to create a table that shows how many states are in each division (using the `division` variable).
 3. When reading in the Excel data set on crime, what do the arguments `sheet=1` and `col_names=TRUE` mean?
 4. Read in the `countyComplete` data set. Use an appropriate function to determine how many observations and variables are contained in the `countyComplete` data set. Write the answer in a complete sentence.

Part II: Improving your plots

1. Examine the relationship between math (`math`) and social studies (`socst`) scores using the HS & Beyond (`hsb2`) data set.
 - a. Start with a basic scatterplot. Create a scatterplot showing math score on the y-axis and social studies score on the x-axis.
 - b. Add a trend line. Add a linear trend line to your plot to help clarify the overall relationship between math and social studies scores. Use a linear model line, but hide the confidence band.
 - c. Group by school type. Check for a different relationship between math and social studies scores depending on school type. Color the points and lines by the `schtyp` variable.
 - d. Improve communicability. Now refine the plot so that it clearly communicates results. Add a title, axis and legend labels, and apply a clean theme.
2. Let's explore the distribution of poverty rates in states around the USA.
 - a. Using the `crime` data, create a boxplot with violin plot of the distribution of `poverty` rate on the y axis, `division` on the x axis, and filling the boxes by `region`. Adjust transparency of boxes and violins as necessary.
 - b. These are multiple divisions in a region, so let's facet on `region` so that each region is in it's own panel. Also get rid of the legend since the name of the region is in the panel already.
 - c. Improve readability. Let's try to fix the overlapping of the x axis labels using `coord_flip()` to rotate the plots horizontally.
 - d. Simplify. Since division is nested within region we don't need ALL the levels of division to show up in each panel. Let's "free" our y axis labels by adding `scales="free_y"` to the `facet_wrap`.
 - e. Add the mean to each group.
 - f. Polish. Finish up by adding a plot title, axis labels and a clean theme. Double check your axis labels before you submit - `coord_flip` makes this backwards sometimes.

Challenge Question

Using the `countyComplete` data set, create a scatterplot to investigate the relationship between the proportion of residents in a county earning a bachelors degree (`bachelors`) income (`per_capita_income`) for west coast states only (CA, OR, WA). Include a linear trend line.

In your visualization, counties should be colored according to the percentage of residents who speak a foreign language at home (`foreign_spoken_at_home`). This variable should be binned into the following ranges before plotting: <10%, 10-20%, 20%+ (*hint: use `case_when`*).

Present the results using separate panels for each state, apply a color scale that is appropriate for categorical groupings, and format the plot so that it is publication-ready with a clear title, labeled axes, and an informative legend.